Recording rules for round 2

For Emranul and Marshal, to try on Saturday 14.11.26. Both record by the same rules, so that the robot learns one way of pushing and not two. This is a first version: the first eight episodes show which rule is hard to keep, and then the rules are changed, not you.

Why: on 11.11 and 12.11 the robot stopped 1 to 2 cm beside the end of the line and drifted to the edge of the line in bends. It copies what the recordings show.

Before each sheet

1. Lay the sheet as the picture on the recording page shows: the round start of the line (end A) on the tape mark, the first straight piece along the taped line. Tape the sheet at its corners, away from the line.
2. The sheet stays taped until all its episodes are done. The page offers them one after the other.
3. Put the puck on the round start of the line, in the middle.

While you guide the arm (motors off)

4. **Middle of the line.** Keep the middle of the puck on the middle of the line the whole way. The black line shows equally on both sides of the puck.
5. **Push from behind,** slowly and at a steady speed. The block stays behind the puck and touches it.
6. **Sharp bend.** Go round the outside of the bend with the block while the forearm turns. Do not turn on the spot, unless the bend is too tight for anything else.
7. **Loop.** Walk the block round the puck the same way every time, as you did in round 1.
8. **Crossing.** Straight across. **Dashed line.** Exactly like a solid line: do not slow down or stop at a gap.
9. **The end.** Be on the middle of the line 3 cm before the end. Then push straight on until the puck covers the round end of the line completely, and stop there. Never stop in the middle of a line.
10. **Wrist.** Keep the wrist as it is at the start. Do not bend it on the way.

In the replay (the robot repeats your path)

11. Watch the puck. If it leaves the line: pause, press **h**, guide it back to the middle of the line and carry on to the end. Record the episode again only when the puck was lost completely.

The score

12. Score 3 only when the puck stayed in the middle all the way and covers the end. Score 2 when it wobbled but stayed on the line. Score 1 for anything else. Be strict: a strict score helps more than a kind one.

After the first eight episodes: for every rule, easy, hard, or not possible. Rules 6, 7 and 10 are the ones most likely to change.

Before Saturday's recording

What is still to do before 14.11.26, and who does it.

Done	Who	What
[ ]	Emranul	Write each printed page's name and an A at the start end on its back, from the list on pages 2 and 3 of the print file.
[ ]	Emranul	Look at the printed lines: solid black, no white stripes. A striped page is printed again.
[ ]	Emranul	Send this page to Marshal. Both read the rules once before Saturday.
[ ]	Emranul and Marshal	Agree on the time, and on who brings what.
[ ]	Emranul	Ask Prof. Nierhoff for fresh copies or the print file of the six course sheets (worth it, not a must).
[ ]	Claude	Check the new recording page: the 16 new tracks, one sheet at a time, the faster arm moves. Send the line that starts it.
[ ]	Claude	Send the numbers of the 36 test runs of 12.11.26.
[ ]	Claude	Send Saturday's checklist, with where the start mark is taped on the table.

Bring on Saturday

The 32 track pages (T01 to T16, solid and dashed), the six course sheets (copies and originals), the red puck, a ruler with millimetres, clear tape, a pen and paper, the laptop with its charger, the cable to the robot.

First things on Saturday, in this order

1. Tape the start mark onto the table once, as Saturday's checklist shows.
2. One watched episode with the faster arm moves: a hand near the stop button. If it looks wrong, one setting brings the old speed back.
3. The first eight episodes: T01 to T04 on the solid prints, each recorded once by Emranul and once by Marshal.
4. Each of you says for every rule: easy, hard, or not possible. Then the rules are changed where needed.
5. Then the other sheets, one sheet at a time, in the order the page offers them.

Always: motors off before anyone touches the arm; never push on a held arm; only one program connected to the robot; the camera and its focus ring are not touched.